

MATTER

Kinetic energy: particles of matter are moving continuously.

* temp $\uparrow$ - K.E of particle $\uparrow$

*

panch tatva

Air earth fire sky water

* movement of particles
gas > liquid > solid

state of matters

- plasma - 4th
- Bose-Einstein condens - 5th

the Gas Which used in cooking cylinder.

→ liquefied petroleum gas (LPG)

* Gas in balloons.

helium hydrogen.

evaporation increases with

- Increase of Surface Area
- Increase of temp.
- Increase in wind speed
- Decrease in humidity.

melting point

minim

which

melts

liquid.

- solid $\xrightarrow{\Delta}$ liquid

Ice $\xrightarrow{m.p.}$ 273.15K

$K = 273 + C$

$C = K - 273$

water boil

100°C

373 K

water freeze

0°C

273 K

* Relation betⁿ fah. & celscius.

$$F = \left(\frac{9}{5} \times C\right) + 32$$

Q. At what temp. the celsius & fahr. scale are same?

→ ① 40°C

$$F = \left(\frac{9}{5} \times C\right) + 32$$

✓ ② -40°C

$$F = \frac{9}{5} \times -40 + 32$$

③ 40K

④ 40F

$$F = -72 + 32$$

$$F = -40^{\circ}\text{C}$$

$$\# -40^{\circ}\text{C} = -40\text{F}$$

$$\frac{C}{5} = \frac{F-32}{9} = \frac{K-273}{5}$$

Q. convert 50°C into kelvin

$$C = K - 273$$

$$50 = K - 273$$

$$K = 323\text{K}$$

$$K = C + 273$$

$$C = K - 273$$

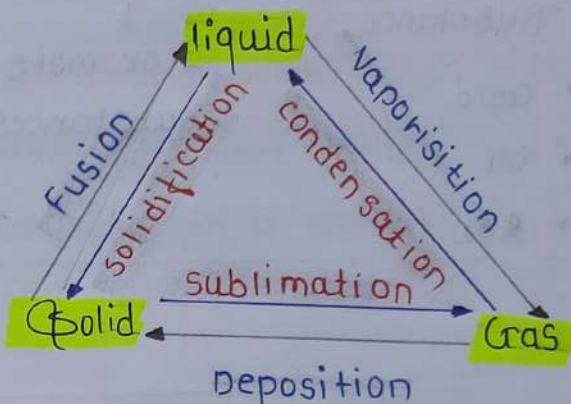
$$F = \frac{9}{5} \times C + 32$$

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

#

$$1 \text{ atmosphere} = 1.01 \times 10^5 \text{ pascal}$$

*



- S - L → Fusion
- L - G → Vaporisation
- G - S → Deposition
- S - G → Sublimation (dry ice)
- G - L → Condensation
- L - S → Solidification

example of Sublimation

- ① camphor
- ② Naphthalene
- ③ Ammonium chloride

example of Deposition

- ① dry ice →
(Which is frozen from CO_2)

MIXTURE

* pure substance

- Gold
- Cu
- O_2

mixture
↓
2 or more substances.

Q. If the door of refrigerator remains open the room's temp. will **increase**.

Matter

pure substance

Elements

compounds

- cannot be broken down to simpler substances.

- copper
- oxygen
- Iron
- hydrogen
- mercury

- Have fixed composition can be broken down into elements by chemicals or electrochemical reaction.

- water
- methane
- sugar
- salt

[No fixed composition]

Homogeneous

(uniform composition)

- sugar in water
- salt in water
- sulphur in carbon disulphide, ~~water~~
- Water in Alcohol

Heterogeneous

(Non-uniform composition)

- Sand & salt
- sugar & salt
- water in oil.

Solution

- homogenous mixture
- solⁿ: solid & liquid & gases CO₂ gas
 - Alloys
 - Iodine in Alcohol
- Alloys are homogenous mixture
- solⁿ of iodine in alcohol known as **tincture of iodine** - Antiseptic solⁿ
 - conc. very low.
 - Iodine (solid)
 - Alcohol (liquid)
- Do not scatter beam of light. (because size of particle very small)
- path of light is not visible
- can't be separated physical.
- cannot seen by naked eyes.

Alloys

- brass - Zn + Cu
- Bronze - Cu + Sn
- stainless steel - Fe + Cr + Ni
- Solder - Pb + Sn
- German silver - Cu + Zn + Ni
- Gunmetal - Cu + Zn + Sn
- cast Iron - Fe + Carbon

• 18K Gold - Ag + Cu + Au

• magnesium - Mg + Al

Saturated solution:

- When no more solute can be dissolved in a solⁿ at a given temp.

Solubility

max. Amount of a solute that can dissolve in a solvent

Amount of the solute present in the saturated solⁿ at this temp.

* unsaturated solⁿ

Amount of solute contained in solⁿ is less than the saturation level.

i) mass by mass% of solⁿ = $\frac{\text{mass of solute}}{\text{mass of soln}} \times 100$

ii) mass by volume% of solⁿ = $\frac{\text{mass of solute}}{\text{volume of soln}} \times 100$

iii) volume by volume% of a soln = $\frac{\text{volume of solute}}{\text{volume of soln}} \times 100$

g. solⁿ contain 40g of common salt in 320 g of water. calculate the concentration in terms of mass by mass % of the solⁿ.

(m) of solute (salt) = 40 g

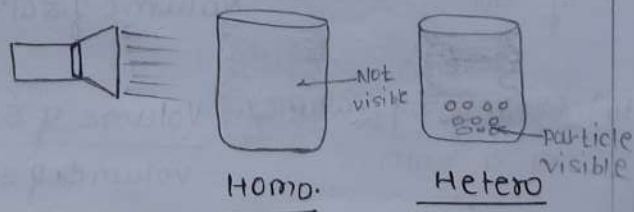
(m) of solⁿ = 320 g

$$\begin{aligned}\text{mass of sol}^n &= (\text{m}) \text{ of solute} + \text{solvent} \\ &= 40 + 320 \\ &= 360 \text{ g}\end{aligned}$$

$$\begin{aligned}\text{mass \% of sol}^n &= \frac{\text{mass of solute}}{\text{mass of sol}^n} \times 100 \\ &= \frac{40}{360} \times 100 \\ &= \frac{100}{9} = 11.1 \%\end{aligned}$$

Suspension

- i) Heterogeneous mixture
- ii) Naked eyes visible
- iii) can be separated by filtration
- iv) Scatter beam of light.

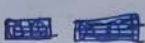


Colloidal Solution

i) heterogeneous mixture

Ex: milk

ii) can't see naked eyes

iii)  scatter beam of light

iv) can't separated by filtration.

v) centrifugation - used separate colloidal particle.

#

Tyndall effect

→ Scattering of a beam of light. is.

- starch solⁿ.
- solⁿ of copper sulphate does not show T.E

- mixture of H₂O & milk show T.E

→ light enters a room through a small hole.

* inside bulb - Neon gas

Intermediate properties b/w those of metals & non-metals they are metalloids.

- boron
- silicon
- germanium

Solution

- Homogeneous
- No filtration
- Not visible to naked eyes.
- Do not scatter beam of light

Ex: Water

colloidal

- Heterogeneous
- No filtration
- Not visible to naked eyes
- scatter beam of light

Ex: milk

Separate $\rightarrow$ centrifugation

Suspension

- Heterogeneous
- filtration ✓
- visible to naked eyes.
- scatter beam of light.

Ex: Flour & water.

Dispersed phase	Dispersing medium	Type	Example
Liquid	Gas	Aerosol	Fog, clouds, mist
Solid	Gas	Aerosol	Smoke, automobile
Gas	Liquid	Foam	Shaving cream
Liquid	Liquid	Emulsion	Milk, Face cream
Solid	Liquid	Sol	Milk of magnesia, mud, pumice
Gas	Solid	Foam	Foam, rubber, sponge
Liquid	Solid	Gel	Jelly, cheese, butter
Solid	Solid	Solid sol	coloured gemstone, milk glass

* Sterling silver

+
Ag + Cu

* separate volatile component (solvent) from its non-volatile solute by the method of evaporation

Centrifugation

- used in blood & urine tests.
- used to separate butter from cream.
- used in washing machines to squeeze out water from wet clothes.

Sublimation

- solid $\rightarrow$ gas
- Naphthalene
- camphor
- Anthracene
- Ammonium chloride

Chromatography

- To separate
 - colours in a dye.
 - pigments from natural colours.
 - drugs from blood.
- * separation of those solutes that dissolve in same solvent.

Distillation

- mixture containing two miscible liquid.

- Ex: gasoline

distilled water

xylene

Alcohol

paraffin

kerosene.

Separating funnel

- mixture containing two immiscible liquid

- Ex: oil & water

Fractional Distillation

- separate Air (used)

Crystallisation

- used to purify solids.

- sea water - salt

↑
remove impurities.

- Separation of crystals of alum from impure samples.

* Chemical + Physical change

- Burning of a candle
- Digestion of Food

* Chemical changes

- Growth of a plant
- Rusting of Iron
- cooking of Food

* Physical changes

- Freezing of water
- Mixing of Iron filing & sand
- melting of ice.
- boiling water.

* example of compound

- sugar
- Table salt (NaCl)
- Water (H_2O)
- CO_2
- sodium bicarbonate
- calcium carbonate

* pure substances

- gold
- copper

compound

- O_2
- Cl_2
- diamond

- water
- salt
- crystals
- baking soda.
- Ice
- calcium oxide
- Iron
- HCl.

example of mixture

- soil
- sea water
- Air
- milk
- Ink (water + dye)
- coal
- soda water
- Brick
- wood

Atoms & molecules

■ Maharishi Kand $\rightarrow$ (Parmanu)

↓
smallest particles beyond which further division will not be possible.

■ pakudha katya yama

↓
particles normally exist in a combined form which gives us various forms of matters.

■ Democritus & Leucippus

↓
When particle obtained can not be divided further.

■ Democritus

↓
Indivisible particles atoms

■ Antoine L. Lavoisier

↓
2 Rules

1] Law of conservation of mass

↓
mass can neither be created nor destroyed in chemical reaction.

2] Law of constant proportion

↓
elements are always present in definite proportion by mass.



element	Atomic No.	Atomic mass
Hydrogen	1	1
Helium	2	4
Lithium	3	7
Beryllium	4	9
Boron	5	11
Carbon	6	12
Nitrogen	7	14
Oxygen	8	16
Fluorine	9	19
Neon	10	20
Sodium	11	23
Magnesium	12	24
Aluminium	13	27
Silicon	14	28
Phosphorus	15	31
Sulphur	16	32
Chlorine	17	35.5
Argon	18	40
Potassium	19	39
Calcium	20	40

- Atomic radius measured in nanometers.

Atom (radius) $\rightarrow 10^{-9} \text{ m}$

- Dalton** $\rightarrow$ 1st use symbols.

 Hydrogen

 Carbon

 Oxygen

 Phosphorus

 Sulphur

 Mercury

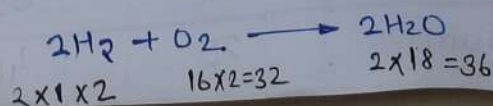
Atomicity

non-metal

Name	Atomicity
Argon	monoatomic
Helium	monoatomic
Oxygen	Diatomic
Hydrogen	Diatomic
Nitrogen	Diatomic
Chlorine	Diatomic
Phosphorus	Tetra-Atomic
Sulphur	poly atomic

- 4 units of hydrogen molecule combines with 32 units of oxygen to give 36 of water mole.

$\rightarrow 36$



Avagadro Number

$$6.022 \times 10^{23}$$

$$60.22 \times 10^{22}$$

$$602.2 \times 10^{21}$$

$$0.6022 \times 10^{24}$$

* 1 mole of oxygen

$$\begin{array}{l}
 \text{atom} = 6.022 \times 10^{23} \text{ atoms} \\
 \downarrow \\
 16 \text{ gm of O}_2 \quad \text{of oxygen}
 \end{array}$$

- 1 mole of anything = 1 gm of that particle thing.

$$\begin{array}{l}
 6.022 \times 10^{23} \text{ atoms of C} \leftarrow 1 \text{ mole of carbon Atom} \\
 \rightarrow \text{12g of carbon}
 \end{array}$$

$$\text{No. of moles} = \frac{\text{given mass}}{\text{molar mass}}$$

$$\text{No of moles} = \frac{\text{No. of particle}}{\text{Avagadro no.}}$$

$$\frac{\text{given mass}}{\text{molar mass}} = \frac{\text{No. of particle}}{\text{Avag. NO.}}$$

Q. find the no. of mole in 52g of Helium (He).

$$\begin{aligned}\rightarrow \text{no. of mole} &= \frac{\text{given mass}}{\text{molar mass}} \\ &= \frac{52}{4} \\ &= \underline{\underline{13}}\end{aligned}$$

Q. 0.5 mole of N_2 gas mass from mole of molecule.

$$\begin{aligned}\rightarrow N &\rightarrow 14 \\ N_2 &\rightarrow 14 \times 2 = 28 \\ \text{no. of mole} &= \frac{0.5}{28} = 14 \text{ g.}\end{aligned}$$

Q. molecular mass of HNO_3

$$\begin{aligned}\rightarrow HNO_3 \\ &= 1 + 14 + 16 \times 3 \\ &= 63 \text{ u.}\end{aligned}$$

Q. No. of particles 46g of Na atoms (No. from mass)

$$\begin{aligned}\rightarrow \text{no. of mole} &= \frac{G.m}{m.m} = \frac{\text{no. of p.}}{A.v. no.} \\ &= \frac{46}{23} = \frac{x}{6.022 \times 10^{23}}\end{aligned}$$

$$x = 12.044 \times 10^{23}$$

Q. How many moles are in 28g of N atom.

$$\begin{aligned}\rightarrow \text{no. of mole} &= \frac{28}{14} \\ &= 2\end{aligned}$$

# 1] electron	- J. J Thomson
2] proton	- Goldstein
3] Neutron	- Chadwick
4] positron	- Anderson

* **J. J Thomson**: 1st one to propose a model for the structure of an atom.

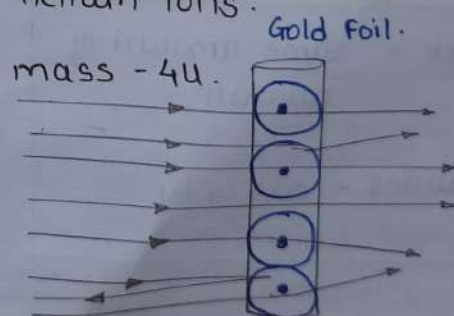
* **Thomson's model**

explained

Atoms are electrically Neutral.

* **Rutherford's model of Atom**

- α -particles are doubly-charged helium ions.



- α -particle scattering experiment result.

- ① passed straight through the gold foil.
- ② some deflected by the foil by small angle.
- ③ out of every 12000 particles appeared to rebound (180° Return).

▪ conclusion

- ① positively charged centre
↓
later called - Nucleus
- ② electrons Revolve around the Nucleus in circular path.
- ③ size of Nucleus very small compared to atom.

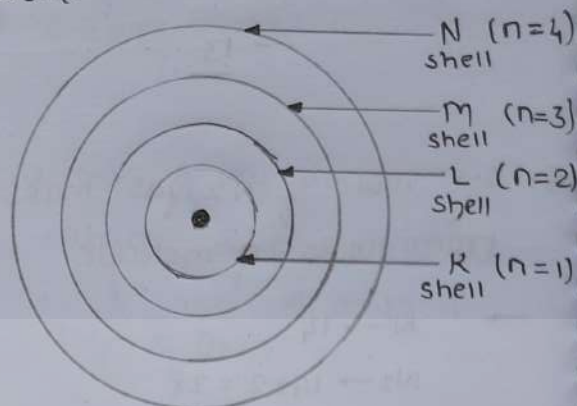
▪ Drawbacks

- ① Revolving electron would lose energy & Fall into the Nucleus.

- Isohyet - same amount of rainfall
- Isohalines - salinity

Bohr's model of Atom

- electrons have specific orbits
- Every orbit has own capacity
- Nucleus very small than the atom.



$$2n^2$$

- K shell - 2 no. of electron
- L shell - 8
- M shell - 18
- N shell - 32

Neutrons

- electron $\rightarrow$ -ve
- proton $\rightarrow$ +ve
- Neutron $\rightarrow$ Neutral

- Neutrons are present in the Nucleus of all atoms.

↓
except

↓
hydrogen

Atomic no. = at. no. of electron

no. of Neutron = mass - proton

Why ~~is~~ Hydrogen - zero.

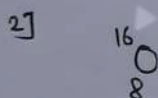


$$x = 1 - 1 \\ = 0$$

$A^* \leftarrow$ atomic mass
 $Z^* \leftarrow$ atomic no.

Ex:

${}^{14}_7\text{N}$
 $\leftarrow$ A. mass
 $\leftarrow$ A. no.



• no. of electron - 8

• no. of proton - 8

• no. of neutron - 8

↓
= mass - proton

$$= 16 - 8$$

$$= 8$$

3] No. of Neutron in Na (sodium)

no. of Neutron = mass - proton

$$= 23 - 11$$

$$= 12$$

Isotope

- same atomic no. but different mass no.

• Ex: Hydrogen

↓
Isotope of (H)

① protium ${}^1_1\text{H}$

② deuterium ${}^2_1\text{H}$

③ tritium ${}^3_1\text{H}$

↓
Radioactive

↓
Application

1] Isotope of uranium used fuel in Nuclear reactors.

2] Isotope of cobalt - treatment of cancer.

3] Isotope of iodine - treatment of goitre.

Isotone

↓
Same no. of Neutron.

Ex: ~~isotone~~

* isotonic solution have same molar concentration.

Isobar

- same mass no. but different atomic no.

• chemical properties diff.

• physical properties same.

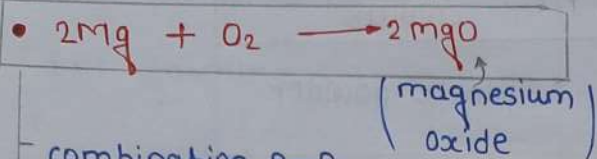
Ex:



Chemical Reactions

Sr. No	compound	colours
1	MgO magnesium oxide	white powder
2	$FeSO_4 \cdot 7H_2O$ ↓ loss of water $FeSO_4$ Ferrous sulphate crystal. (solid)	green ↓ to white
3	Fe_2O_3 Ferric oxide (solid)	Reddish Brown
4	NO_2 Nitrogen oxide (gas)	brown fumes
5	$AgCl$ Silver chloride (solid) ↓ in sunlight	White ↓ grey
6	$CuSO_4$ copper sulphate (solution)	blue
7	$FeSO_4$ Iron sulphate (aq)	blue-green
8	ZnO copper(II) oxide	black
9	PbI_2 lead iodide	yellow
10	$BaSO_4$ Barium sulphate	White powder
11	Litmus soln Saturated compound unsaturated compound	→ purple dye → Blue clean flame → yellowish flame

* magnesium ribbon burns with a dazzling white flame & change into white powder. → this powder



combination Rean

Heat & light are produced.

skeletal chemical equation

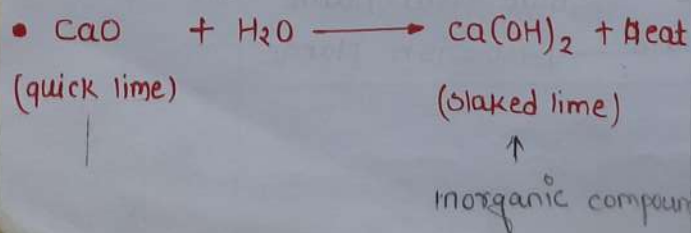
eqn is unbalanced because mass is not the same on both sides of eqn.

Combination Reaction

- $\text{CaO} \rightarrow$ quick lime (calcium oxide)
- $\text{Ca(OH)}_2 \rightarrow$ slaked lime (calcium Hyd.)
- $\text{CaCO}_3 \rightarrow$ calcium carbonate.

→ single product is formed from two or more reactants.

1] calcium oxide react^{with} water produce slaked lime.



slaked lime : used

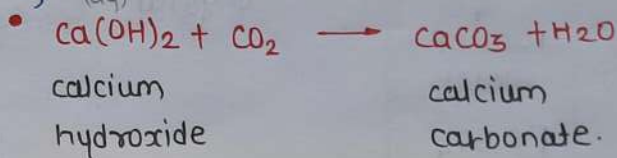
↓
inorganic compound

↓
White washing wall.

CaCO_3 (calcium carbonate) : used

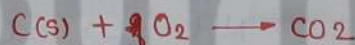
↓
shiny finish to the walls.

2] (aq)

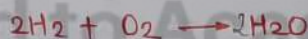


↑
a formula for marble also.

3] Burning of coal



4] formation of water



exothermic reactions

↓
large Amount of heat is evolved.

↓
mixture

(warm)

① Burning of Natural gas



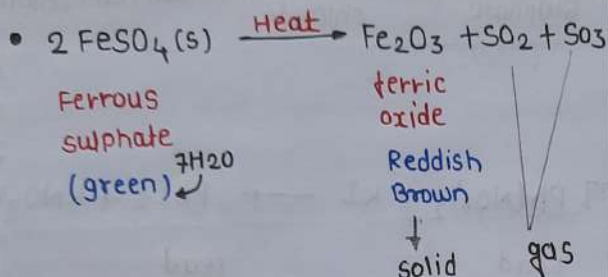
② respiration rean



③ decomposition of vegetable matter into compost.

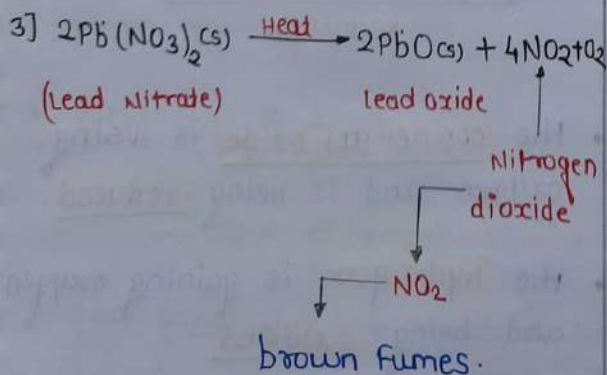
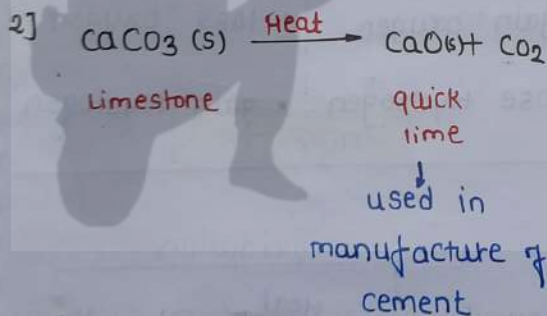
Decomposition Reaction

Single reactant breaks down to give simpler products.

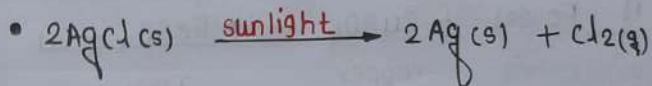


→ Ferrous sulphate crystals $(\text{FeSO}_4 \cdot 7\text{H}_2\text{O})$ (green) lose water

$\xrightarrow{\text{Heat}}$
 colour changes
 ↓
White



4] (white) silver chloride turns grey in sunlight.



↓
 * both reactants used in
black & white photography

* Decomposition Reaction

require energy in form of
 • heat
 • light
 • electricity
 for break down the reactants.

Endothermic Reactions

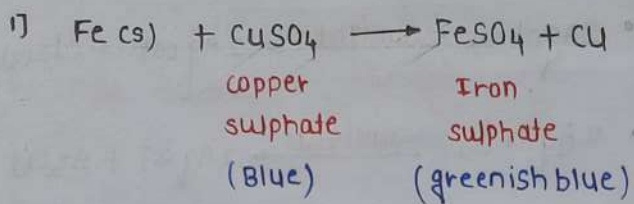
energy is absorbed.

Reactant mixture: cool.

exothermic Reaction endothermic Reaction

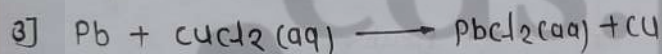
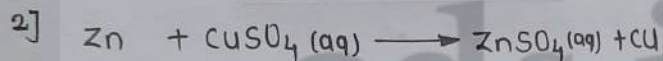
- heat is evolved - Reactant mixture ↓ warm	- heat is absorbed - Reactant mixture ↓ cool
--	---

Displacement Reaction



• $\text{Fe} > \text{Cu}$

↳ Iron has displaced (or) Removed another element, copper from copper sulphate soln.

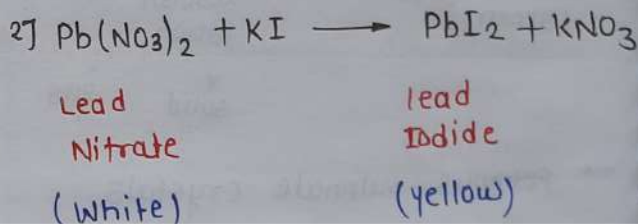
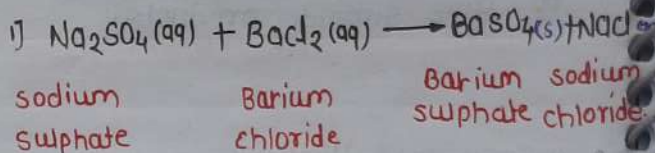


↳ zinc & lead are more reactive than copper.

Reactivity series

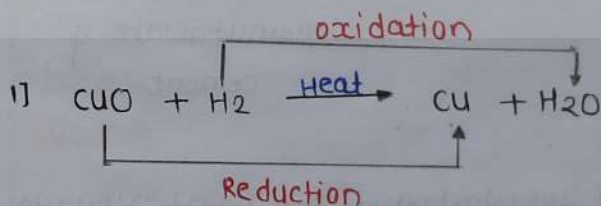
- K
- Na
- Ca
- } cold water
- Mg
- hot water
- Al
- Zn
- Fe
- } steam
- Pb
- Cu
- Hg
- Ag
- Au
- } not react with water

Double Displacement



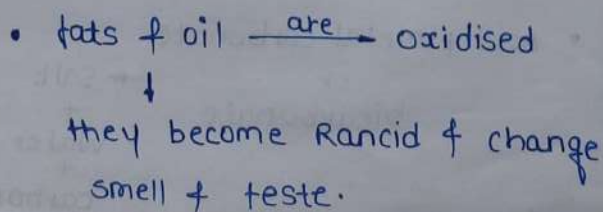
Oxidation + Reduction = Redox

oxidation	Reduction
• gain oxygen	• loss oxygen
• lose Hydrogen	• gain Hydrogen



- the copper (II) oxide is losing oxygen and is being reduced.
- the hydrogen is gaining oxygen and being oxidised.

• copper (II) oxide → Black.



- ⑧ **Nitrogen** → to prevent the chips from getting oxidised

Acid, Base & salts

Acids

- sour in taste
- blue litmus to red

Bases

- bitter
- Red litmus to blue

Litmus

- Litmus solution - purple dye
- extracted - lichen
↓
belonging to
Thallophyta.

Natural Indicator

another

Natural material

Indicator	Acid	Base
turmeric (yellow)	yellow	Red

Synthetic Indicator

Indicator	Acid	Base
phenolphthalein (colourless)	colourless	pink
Methyl orange (orange)	Red	yellow

olfactory Indicator

- smell / odour

Indicator	Acid	Base
onion	does not change its smell	loses its smell
clove oil	✓	x
vanilla	does not vanish with Acid	vanish with base

red cabbage leaves

turmeric

coloured petals of some flowers

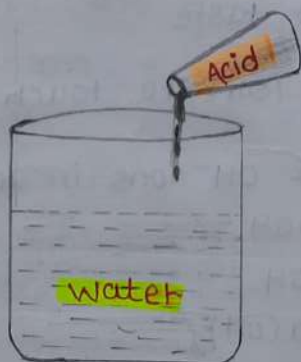
- Hydrangea
- petunia
- Geranium

- Acid + Metal $\rightarrow$ salt + H_2
- Base + Acid $\rightarrow$ salt + water
- metal oxide + Acid $\rightarrow$ salt + H_2O
- Acid + metal carbonate or bicarbonate $\rightarrow$ salt + water + carbon

Acid

- Acid have sour taste
- corrosive in Nature
- Release H^+ ions in aqueous soln
- Example:
 - H_2SO_4 (sulphuric Acid)
 - CH_3COOH (Acetic Acid)
 - HNO_3 Nitric Acid.
- The electric current is carried through the Acidic soln by Ions.
- The process of dissolving an Acid or base in water is a highly exothermic one.

↓
The Acid must always be added slowly to water with constant stirring.

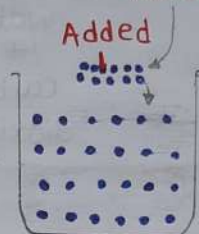


concentrated Acid

- Acids that contain only a small quantity of water.

- Acid $\uparrow$, Water $\downarrow$

more Acid

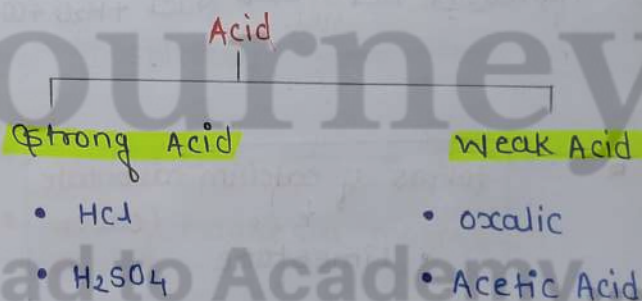
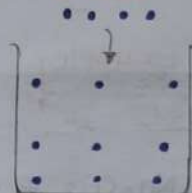


Dilute Acids

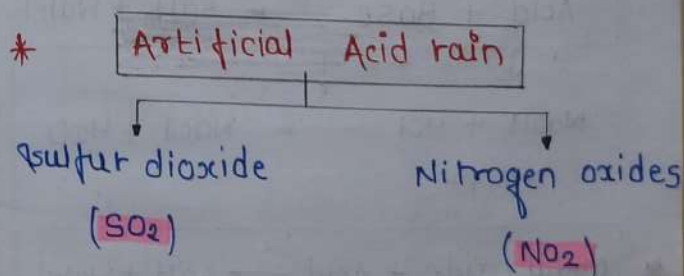
- Acids that contain more water than the C.A.

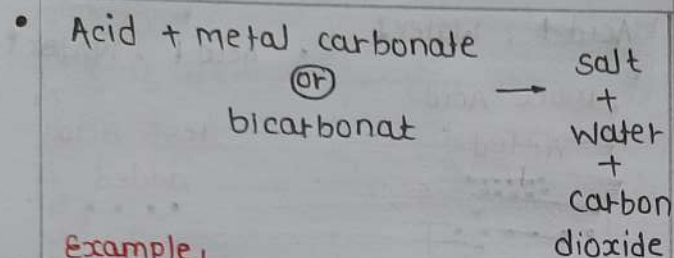
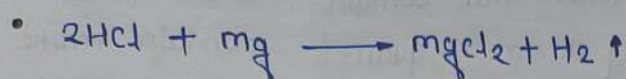
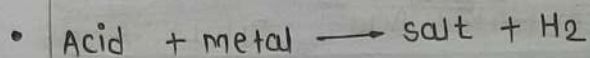
- Acid $\downarrow$, Water $\uparrow$

less Acid added

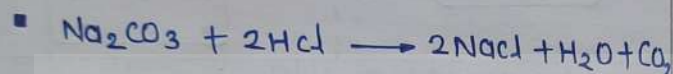


* The atmosphere of **venus** is made up of a thick white and **yellowish clouds** of **sulphuric Acid** (H_2SO_4)





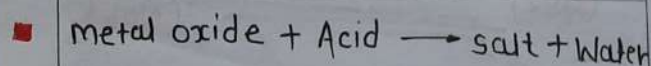
Example $\downarrow$



forms of calcium carbonate
($CaCO_3$)

- Limestone
- chalk
- marble ($CaCO_3$)

Neutralization Reaction

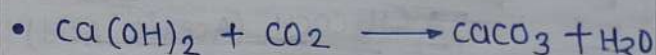
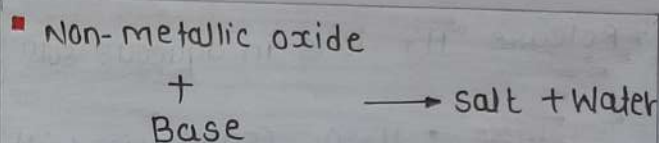


metallic oxide

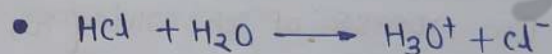
basic oxide

Non-metallic oxide

Acidic oxide



* the separation of H^+ ion from HCl molecules cannot occur in the absence of water.



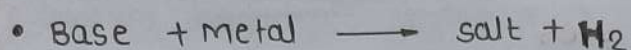
glucose + Alcohol soln do not conduct electricity.

Bases

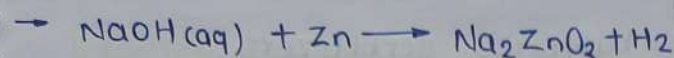
- bitter taste
- soapy / soft to touch
- Release OH^- ions in aqueous soln
 - $NaOH$
 - KOH
 - $Ca(OH)_2$

- Alkalies : Water soluble bases.

- All bases do not dissolve in water.

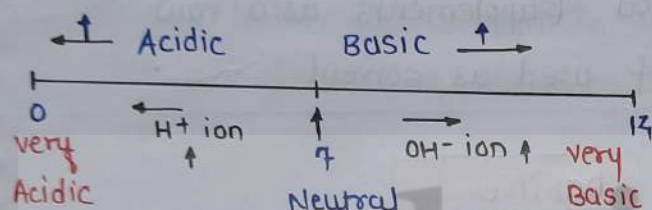


↑
soluble in
water.



sodium
zincate

PH scale



	PH
HCl	0
Gastric Juice	1.2
Lemon Juice	2
Apple	3
Tomato Juice	4.5
COFFE	5-6
carrot Juice	6
saliva (after meal)	5
Banana	4
milk	6-7
body PH	7.0-7.8
Blood	7.4
Bile	8.2
pancreatic Juice	8.6
Litmas soln	7
milk of magnesia	10
NaOH	14

* When pH of rain water

less than 5.6

Acid rain

* Tooth decay starts when the PH of mouth is lower than 5.5

tooth enamel made up of

calcium hydroxyapatite

(crystalline form of calcium phosphate).

■ Toothpaste $\longrightarrow$ Basic.

■ milk of magnesia (magnesium hydroxide)

Antacids.

Acids	Source
Lactic Acid	curd, milk
Ascorbic Acid	Amla
oxalic Acid	Tomatoes
citric Acid	orange, Lemons
malic Acid	Apple
Tartaric Acid	Grapes, mango
	Tamarind
Formic Acid/	Ant's sting
methanoic Acid	Nettle sting
Acetic Acid	Vinegar.

Acids	uses
Hydrochloric Acid	Remove rust from metals
Sulphuric Acid	automobile batteries fertilizers, paints plastic & dyes
Nitric Acid	used in fertilizers, plastics photographic films explosive & dyes.
Acetic Acid	used in food supplements as a raw material & used as solvent.

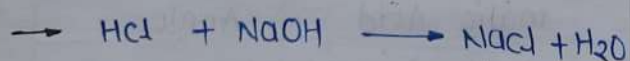
Salt

■ salt of strong Acid + strong base $\rightarrow$ Neutral (PH-7)

■ salt of strong Acid + weak base $\rightarrow$ Acidic (PH- $\downarrow$ 7)

■ strong base + weak Acid $\rightarrow$ basic (PH- $\uparrow$ 7)

■ salt formed by



Neutral salt

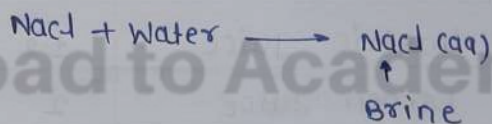
common salt

■ symbol struggle for freedom

NaCl

brine

- When electricity is passed through an aqueous soln of sodium chloride.

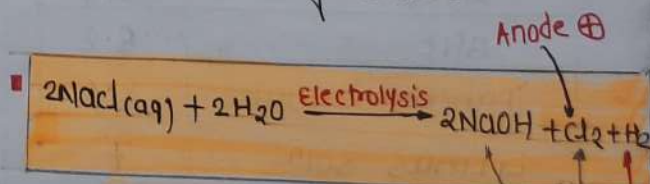


Chlor-Alkali process

aqueous soln of sodium chloride (called brine)

decomposes to form

Sodium hydroxide



Anode $\oplus$

used in manufacture of bleaching powder.

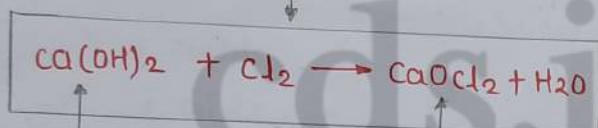
cathode $\ominus$

- Sodium hydroxide solution is formed near the cathode.



chlorine is produced during electrolysis of brine

This chlorine gas is used to for manufacture of bleaching powder.

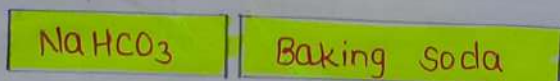


↑
dry slake
lime

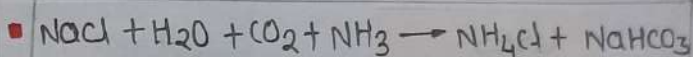
↑
calcium
Hypochloride

used

- linen in the textile industry
- paper factories
- washed clothes in laundry
- oxidising agent
- make drinking water free from germs.

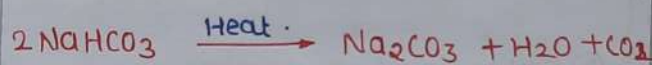


- Sodium hydrogen carbonate
- produced : using Sodium chloride (NaCl)
- Antacids. (8-9 Ph)



■ NH₄Cl - Ammonium chloride

■ NaHCO₃ - sodium hydrogencarbonate.



↑
sodium
carbonate

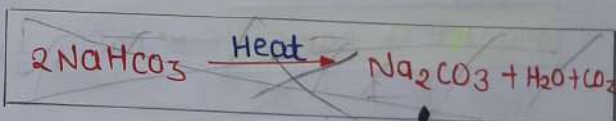
uses

- ① Antacid (milk of magnesia also used Antacid)
- ② soda - Acid fire extinguishers
- ③



Washing soda

- Sodium carbonate
- obtained from : sodium chloride



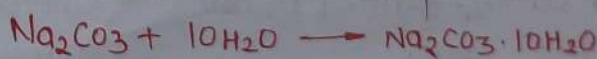
↑
Washing
soda

(sodium carbonate)

↑
basic salt

recrystallisation
of sodium
carbonate

gives



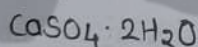
USES

- ① used in glass, soap & paper industries.
- ② manufacture of sodium compound such as borax.
- ③ cleaning agent
- ④ removing permanent hardness of water.

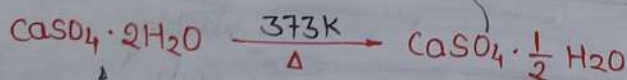
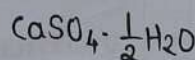
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ - hydrated copper sulphate

- Five water molecule present in one formula unit of (CuSO_4) .

gypsum

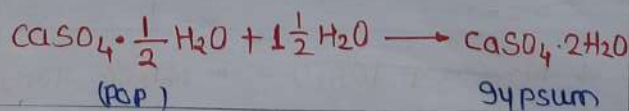


plaster of paris



gypsum

calcium sulphate hemihydrate.



• blue vitriol - CuSO_4

Metals & non-Metals

basic-oxides

Metals

lustre

metals in their pure state

have a shining surface

malleability

beaten into thin sheets

most malleable metals

- Gold
- silver

sonorous

- metals that produce a sound on striking a hard surface.

- school bells are made

ductility

• Drawn into thin wires

1 gm of gold
↓
about 2 km wire

- most ductile
- Gold

good conductors of heat

best conductors of heat

- silver
- copper

poor conductors of heat

- lead
- mercury

High melting points

but • gallium,
• caesium
low m.p

hard

solid at room temp.

except

mercury — liquid at room temp.

Non - Metals

Non lustrous

except.

- Iodine: Non metal but lustrous.

Carbon - non metal

Allotrope

Diamond

- hardest sub.
- very high melting + boiling point

Graphite

- Good conductor of electricity

Solid & Gas

except

Bromine

liquid at room temp.

produce Acidic oxides

When dissolve in water.

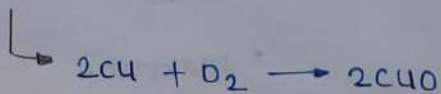
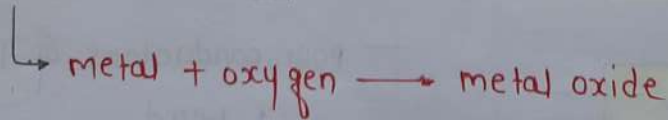
Alkali Metals

- Lithium
 - Sodium
 - Potassium
- low densities + low melting point.

↓
Soft & they can cut with knife

chemical properties

① metal in air

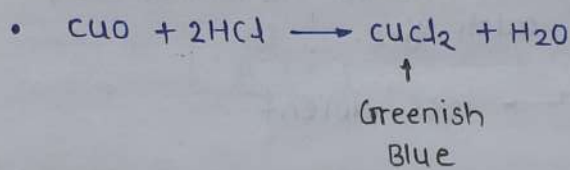


copper(II) oxide
(black)

⊕ Anodising

↓
Al

- metal oxide + Acid $\rightarrow$ salt + H_2O
(basic)



- metal oxide

Basic

- Non metal oxide

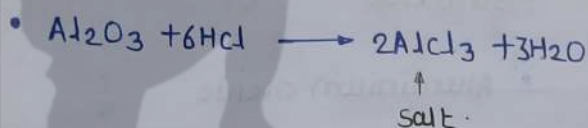
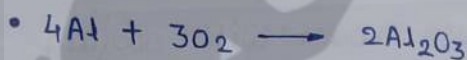
Acidic

both Acidic & Basic

Amphoteric oxides

Aluminium
oxide

zinc oxide



- Most metal oxide are insoluble in water.

except

potassium
oxide

Sodium
oxide

Dissolve in water to produce Alkalies.

- Silver & Gold

do not react with oxygen.

- React so vigorously with O_2

they catch fire if
kept in open

- potassium
- sodium

they are kept immersed in kerosene oil.

Anodising

forming thick oxide layer of Aluminium.

Aluminium article is made the Anode & electrolysed with dilute sulphuric acid.

K ■ potassium
 Na ■ sodium
 Ca ■ calcium

react with cold water } reaction violent & exothermic that evolved H_2 immediately catch fire.

cold water → less violent

Mg ■ magnesium → React with hot water

Al ■ Aluminium

Zn ■ zinc

Fe ■ Iron

react with steam

Pb ■ Lead

Cu ■ copper

Hg ■ mercury

Ag ■ silver

Au ■ Gold

Do not react with water at all.

* metals Float When treated with water

calcium

(cold)

magnesium

(Hot)

bubbles of H_2 gas sticking to its surface.

Ag + sulphure → Black colour.

oxides react with sodium hydroxide.

• Aluminium oxide

• zinc oxide

• silicon dioxide

calcium oxide NOT react with NaOH. basic

• Basic

→ metallic oxide does not (Basic in Nature) react with base.

Non metal + O_2 → Non metal oxide
↓
Acidic

■ When metal reacts with Acid

H₂ gas is evolved

But

■ metal react with Nitric Acid [HNO₃]

H₂ gas is not evolved

because → HNO₃ is strong oxidising agent.

Reduction
↓
- gaining of H
- losing of O₂

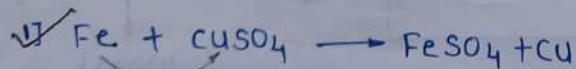
But 😊

magnesium (mg) & manganese (mn)

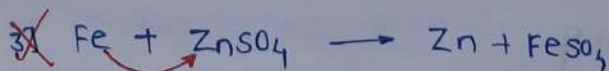
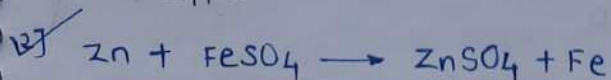
react with very dilute HNO₃ to evolve H₂ gas.

■ metal + Acid → salt + H₂

■ Redⁿ is feasible.



Iron more reactive than copper.

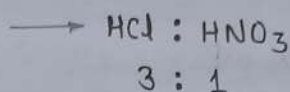


Iron less reactive than zinc

Aqua regia

freshly prepared mixture of concentrated hydrochloric acid + conc. Nitric Acid in ratio of

3:1



- highly corrosive fuming liquid
- It is one of the few reagents that is able to dissolve gold & platinum.

Ionic comp.

- transfer of electrons
- form metal to non-metal.

covalent comp.

- sharing of electron
- insoluble in water.

Ionic compounds

- solids
- hard because strong force of Attraction $\oplus \& \ominus$ ions.
- have high m.p. boiling point.
- also called electrovalent comp.
- generally soluble in water.
- ↓
kerosene, petrol
- ↓
solid $\oplus$ insoluble in solvents
- conduct electricity in molten state

- K
- Na
- Ca
- Mg
- Al

electrolysis of
molten ore

- Zn
- Fe
- Pb

Reduction

- Cu
- Ag
- Au

free state.

Roasting

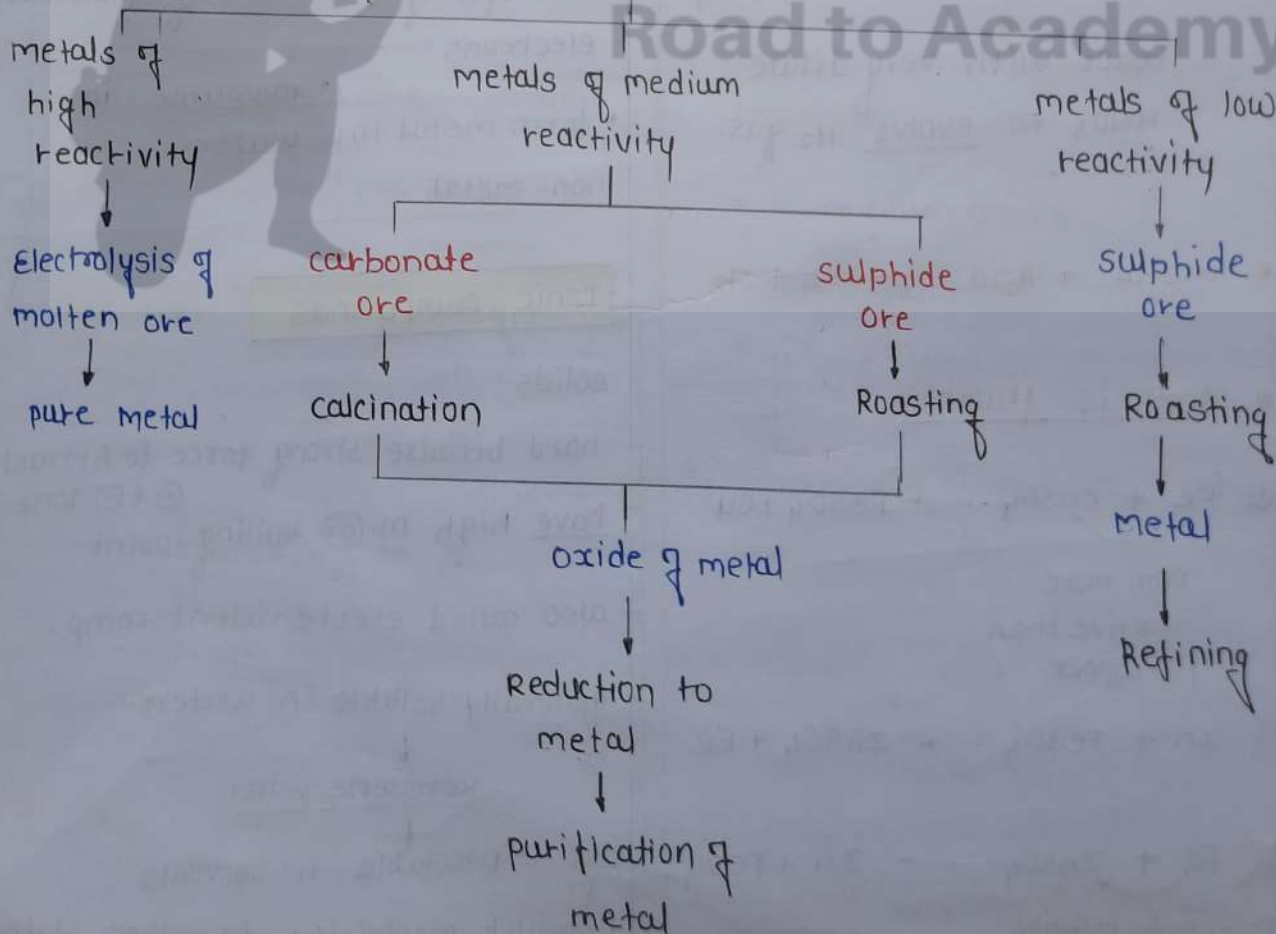
- sulphide ores
are converted
into oxide by
heating strongly
in presence of
excess air.

calcination

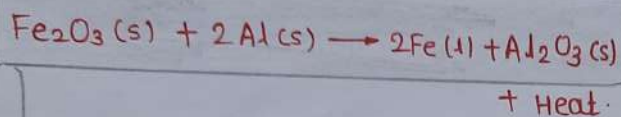
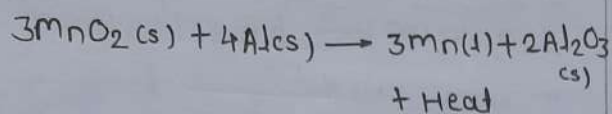
- carbonate ores
are changed
into oxide by
heating strongly
in limited air.

■ mercury $\xrightarrow{\text{ore}}$ cinnabar

concentration of ore



- manganese dioxide is heated with aluminium powder



- These Displacement Reactions are highly exothermic
- Iron (III) oxide (Fe_2O_3) with Aluminium is used to join railway tracks (or) cracked machine part.

- obtained by the electrolysis of their molten chlorides.

- sodium
- magnesium
- calcium

- obtained by the electrolytic Reduction of Aluminium oxide

↓
Aluminium.

■ ores

- pitch Blend → uranium
- Borax → sodium
- Galena → Lead.

corrosion

- silver + sulphur $\xrightarrow{\text{Air}}$ coating
 $\text{Ag} + \text{S} \rightarrow \text{Ag}_2\text{S}$ (black) silver sulphide (black)
- Copper + CO_2 (moist) → copper carbonate (CuCO_3) (green coating)

prevention of corrosion

- ① Alloys
- ② Anodising
- ③ Galvanizing → thin (coating) layer of zinc
- ④ oiling
- ⑤ painting
- ⑥ greasing.

⑧ Alloys

- stainless steel → Fe + Ni + Cr
- brass → Cu + Zn
- Bronze → Cu + Sn
- solder → Pb + Sn
- Gun metal → Cu + Sn + Zn + Pb

■ Ionic compound

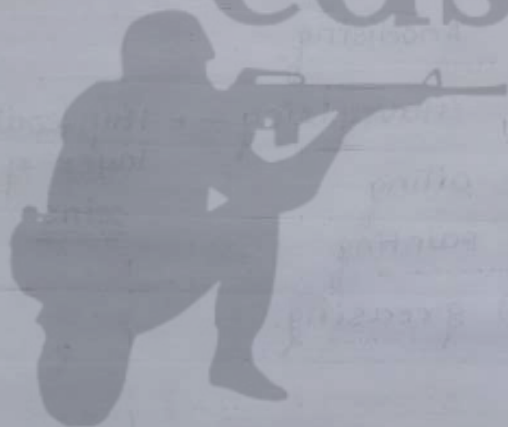
• Boiling point

$\text{CaO} > \text{CaCl}_2 > \text{NaCl} > \text{MgCl}_2 > \text{LiCl}$

• melting point

$\text{CaO} > \text{NaCl} > \text{CaCl}_2 > \text{MgCl}_2 > \text{LiCl}$

cds.journey



Road to Academy...

Carbon & its compound

* The earth crust ^{in the form of} ^{minerals.}

carbon - 0.02%

0.025%

* atmosphere has

CO₂ - 0.03%

Most carbon compound are poor conductors of electricity.

→ atomic No. of carbon - 6

→ valency of carbon - 4

* O=O * H-H

* N≡N.

→ carbon is tetravalent.

→ diamond and graphite are formed by carbon atom.

In Diamond (single bond)

→ each carbon atom is bonded to four other carbon atoms forming three-dimensional structure.

In Graphite - good conductor of electricity

→ double-bond

→ c - c - c → hexagonal.

||
c

→ smooth & slippery

→ each carbon atom is bonded to three other carbon atoms in the same plane.

Fullerenes

→ carbon allotropes.

→ shape of Football (arranged)

→ 60-carbon atoms.

Catenation :

→ carbon has unique ability to form bonds with other atoms of carbon, rise to large molecules.

Single bond

↓
Saturated compound

double / triple bond

↓
unsaturated compound.

→ carbon-carbon bond is very strong & stable.

→ one reason for the formation of strong bonds by carbon is its small size.

→ bond formed by element having large atoms are much weaker.

single bond $\rightarrow$ ane (alkane)

Saturated carbon compound.

* these carbon compounds are normally not very reactive

double bond $\Rightarrow$ ene (alkene)

unsaturated carbon compound.

Triple bond $\equiv$ yne (alkyne)

unsaturated carbon compound.

1) Meth----- 

2) Eth----- 

3) prop----- 

4) But-----

5) pent-----

6) Hex-----

7) Hept-----

8) Oct-----

9) Non-----

10) Dec-----

Formula

① Alkane - C_nH_{2n+2}

② Alkene - C_nH_{2n}

③ Alkyne - C_nH_{2n-2}

unsaturated carbon compound are more reactive than the saturated carbon compound.

Structural isomers

$\rightarrow$ identical molecular formula but different structures are called....

$\rightarrow$ carbon atoms arranged in the form of a ring $\rightarrow$ cyclo

Ethane [C_2H_6]

$\rightarrow$ 7 covalent bonds.

1) Alcohol - OH (-OH)

2) Aldehyde - CHO $\begin{array}{c} \text{H} \\ | \\ -C=O \end{array}$

3) ketone - $C=O$ $\begin{array}{c} \text{O} \\ || \\ -C- \end{array}$

4) carboxylic acid - COOH $\begin{array}{c} \text{O} \\ || \\ -C-OH \end{array}$

Alcohol $\rightarrow$ suffix -ol $\begin{array}{c} | & | & | \\ -C- & -C- & -C- \\ | & | & | \\ & OH & \end{array}$
propanol

Aldehyde $\rightarrow$ suffix -al $\begin{array}{c} | & | & | \\ -C- & -C- & -C=O \\ | & | & | \\ & & \end{array}$
propanal

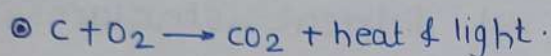
ketone $\rightarrow$ suffix -one $\begin{array}{c} | & | & | \\ -C- & -C- & -C- \\ || & & \\ O & & \end{array}$
propanone

carboxylic acid $\rightarrow$ suffix -oic acid
Acid $\begin{array}{c} | & | & | \\ -C- & -C- & -C=O \\ | & | & | \\ & OH & \end{array}$
propanoic acid.

1) $\text{H}-\overset{\text{H}}{\underset{|}{\text{C}}}=\text{O} \rightarrow \text{Methanal}$

combustion

→ carbon burns in oxygen to give carbon dioxide along with release of heat & light.



→ Saturated hydrocarbon will generally give clean flame.

* # gas - red because lack of O_2

pyq 2022

→ unsaturated carbon give yellow flame with lots of black smoke.

→ the oil & gas seeped into the porous part of the rock.

Imp

→ Animal Fats generally contain Saturated Fatty acid.



harmful for health.

→ oil containing unsaturated fatty acids.

Ethanol

→ Ethanol is liquid at room temperature.

→ good solvent.

→ used medicines such as tincture Iodine, cough, syrups & Tonics

→ consumption of small quantities of dilute ethanol causes drunkenness.

→ Methanal reacts rapidly with the components of cell. It causes the protoplasm to get coagulated.

IMP

→ Methanol affects the optic nerve, causing blindness

→ Euglenaceae plants are one of the most efficient convertors of sunlight into chemical energy.

QUP → ethanol production (TOP)

→ ethanoic acid is commonly called acetic acid.

→ 5-8% soln of acetic acid in water is called vinegar.
used. ←

preservative in pickles.

→ carboxylic acid - weak acid.

Esters

→ Formed by reaction of an acid & an alcohol.

* Ethanoic acid + Ethanol → Ester.

→ sweet-smelling substances.

→ used perfumes, flavouring agents.

Saponification → soap.

Q-1] Butanone is four-carbon compound with the functional group?

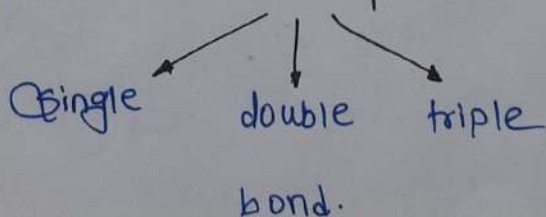
→ ketone

high melting & boiling point

Acetic Acid > chloroform > ethanol > methane

CH_3COOH > CHCl_3 > $\text{CH}_3\text{CH}_2\text{OH}$ > CH_4

carbon atoms may be linked

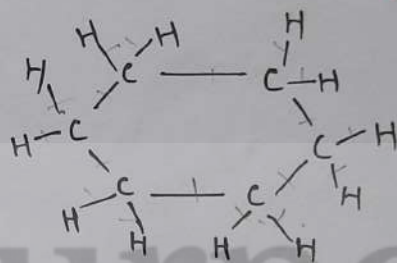


* Silicon forms compounds with hydrogen which have chains of upto seven or eight atoms.

CDS PYQ 2022-I

* No. of saturated and unsaturated bond in cyclohexane.

Structure. formula
↓
 C_6H_{12}

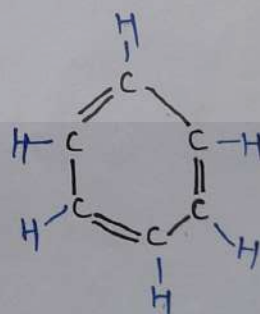


* single bond (saturated) - 18

* double bond (unsaturated) - 0

Imp.

* Benzene with molecular formula C_6H_6



single bond - 9

double bond - 3

* Imp

→ oxygen rich mixture is burnt to give a clean blue flame